

Group Design Project



Name	Roll No.
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Easyfarm

Vertical Farming barrel







Hydroponic Vertical Barrel



Market Study

- A lot of people living in city areas do not have enough area to farm, growing plants in grow bags does not ensures a good yield
- By providing an alternative for easily growing plants by using Hydroponics, a lot of people can benefit from this.
- It is also helpful in areas without fertile soil and even in deserts



Market Study

- Brand new concept : A number hydroponics related products are available in the market but they all are heavy and very costly. Our product is modular.
- Target audience: people who are living city areas where there is a lack of land and fertile soil and water for cultivation.
- Supply: At first the product will be available in agricultural stores in the major cities



Market Study

- Supply: At first the product will be available in agricultural stores in the major cities ,then as the demand increases the supply will be expanded to the tier 2 cities and so on, The product is injection molded. Institutionally funded. Channels of distribution through agencies associated with the funding institution.
- Risks underlying: the initial acceptance in the market , convincing the suppliers



UN sustainable development goals aligned with 'Easyfarm'

Goal 1

End poverty in all its forms

Landless people can cultivate and harvest their essential crops they can also sell the surplus items to generate an income, by a one time investment



UN sustainable development goals aligned with 'Easyfarm'

Goal 2

Zero Hunger

Can curb the hunger of a family easily without much effort and expertize. Also can sell or donate the surplus items to curb others hunger.



UN sustainable development goals aligned with 'Easyfarm'

Goal 3

Health

People can eat healthy food cultivated by themselves without minimal usage of pesticides or insecticides.



UN sustainable development goals aligned with 'Easyfarm'

Goal 11

Cities

Sustainable type agriculture in cities where there is a limitation of land and space.



UN sustainable development goals aligned with 'Easyfarm'

Goal 12

Sustainable consumption and production

Deforestation is one of the major cause of climate change, increasing human population increases the demand of land and agriculture. With the use of easy farm the demand of land can be met to great extent and thereby contributing less to the global climate change.



Yield Comparison

Conventional farming VS Easyfarm Vertical barrel

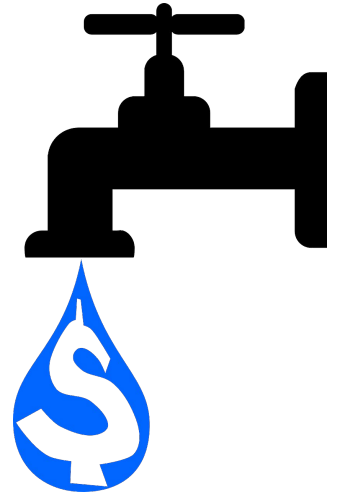
- In conventional farming , **one hectare** of land can yield about **45 quintals** yield of peas.
- One hectare= 247.105cents.
- So one cent can yield $4500\text{kg} / 247.105 = 18.21$ kg of peas .
- Also one cent =404685.64224 sq. cm
- Bucket base diameter of easy farm = 60 cm.
- So base area covered by easy farm module = $A = \pi r^2 = 3.14 * 60 * 60 = 11,304$ sq. cm



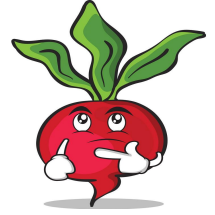
Yield Comparison

Conventional farming VS Easyfarm Vertical barrel

- After analyzing in 1 cent we can place $=404685.64224/11304 = 35.8$ easyfarm barrels.
- Taking in account the space to be provided between the barrels we can place about 25 barrels of 'easyfarm' in one cent.
- In short as 'easyfarm' can give us a yield of about **20 times than that of conventional farming methodology.**



WHAT CAN WE GROW?



- Tomato
- Capsicum
- Lettuce
- Pepper
- Spinach
- Chilly
- Coriander
- Cauliflower
- Beans



- Gourd
- Lady finger
- Corn
- Sesame seeds
- Mint
- Brinjal
- Cucumber
- Basil
- Radish



X- FACTOR CROPS

1. **TOMATO:-** It is an all season vegetable in India. 3rd on the demand list of vegetables in India with 67%. 2nd on the demand list when the whole world is considered. Hence the best crop to be grown in hydroponics system.
2. **LETTUCE:-** It is an all season vegetable in India. 6th on the demand list of vegetables in India with 53%. They have a crop value of 40% in the Indian market.



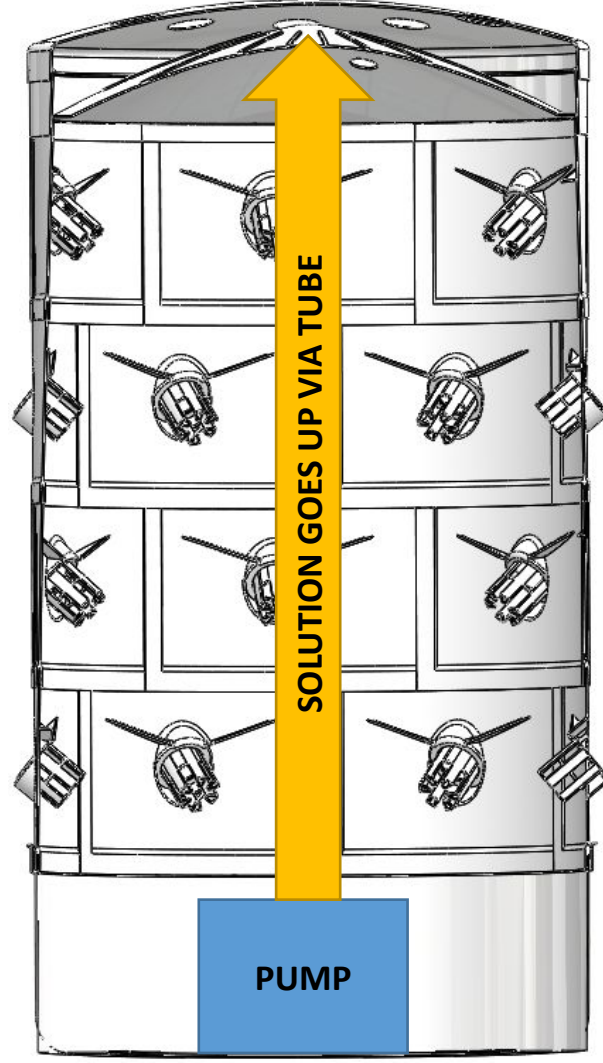
X- FACTOR CROPS

3. **SPINACH:-** It is an all season vegetable in India. 8th on the demand list of vegetables in India with almost 50%. They have a crop value of 40% in the Indian market.
4. **CORIANDER:-** It is an all season vegetable in India. It is a very popular medicinal plant that belongs to Apiaceae family in taxonomic classification, which is widely used as a spice and also in pharmacy and in food industries. Due to its multiple uses it is of really high demand in India.

Working

- Mineral water pumped to the buffer drips down irrigating the plants.
- Gaps are given in the base bucket to prevent leaking water (if any) from reaching the floor.
- Snap fit locks are tapered for ease of assembly

<https://docs.google.com/presentation/d/1DYTqdvVdQ1lhDkFIARyhrXKJcSGtORWTDnu8qPJcBOU/e>
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SOLUTION GOES UP VIA TUBE

PUMP

COMES DOWN IRRIGATING THE PLANTS

Video



**EASY
SNAP IN
ASSEMBLY**

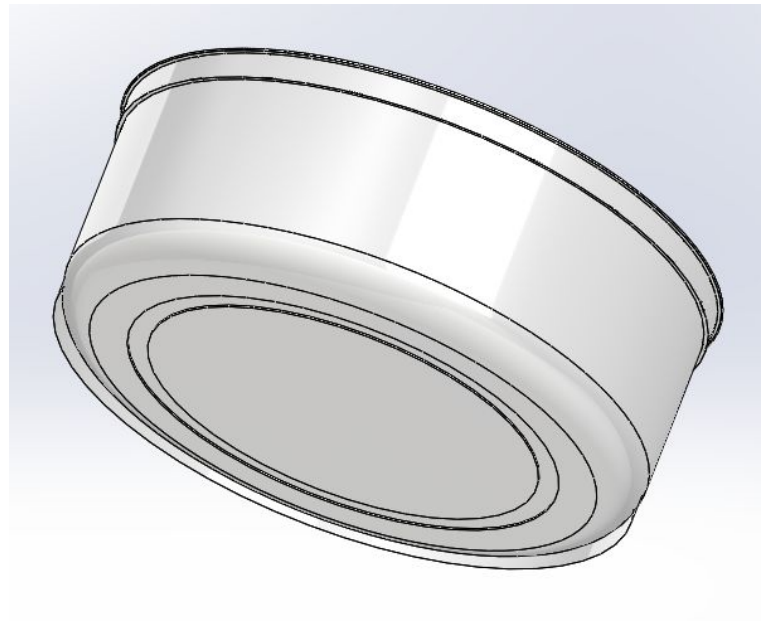
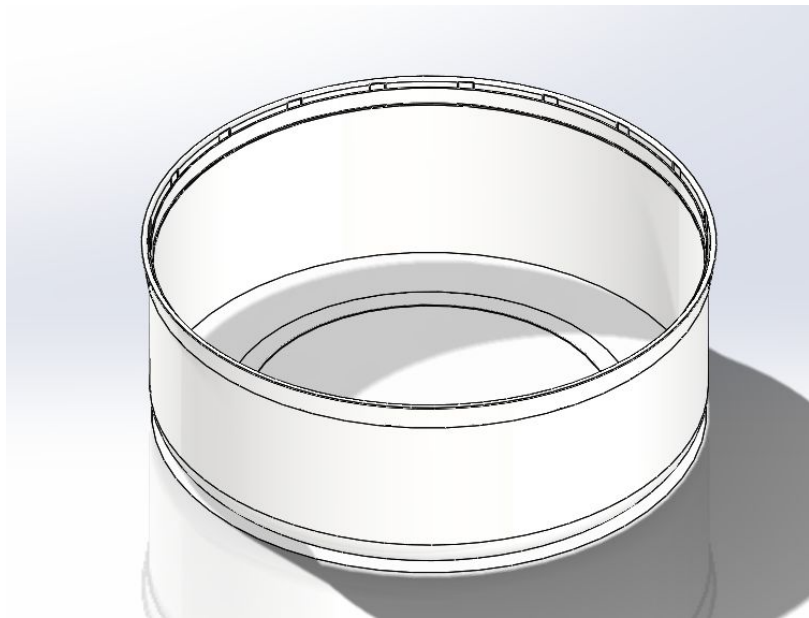




Components



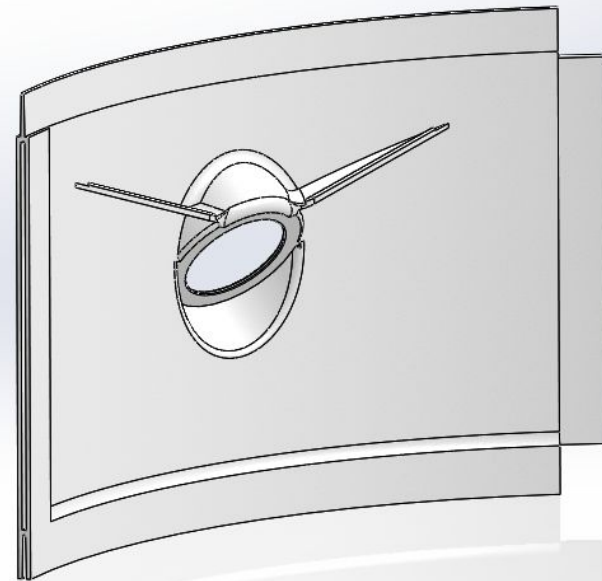
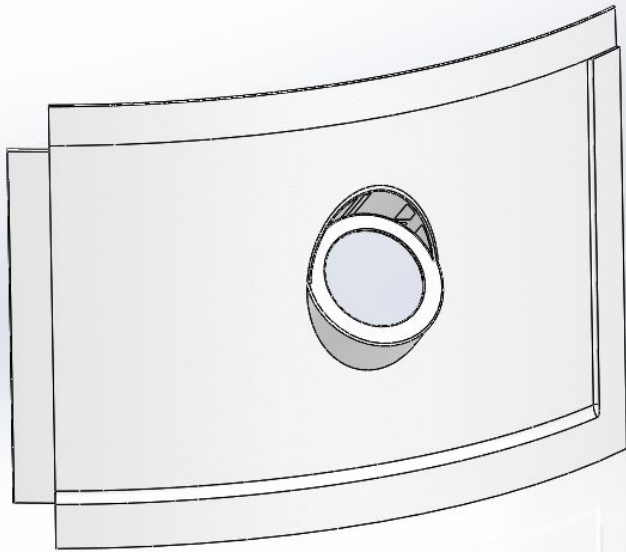
Base



Base

- Capacity = 50 litres
- Material = PVC Sheet
- Mass = 1123 grams
- Manufacturing method = Vacuum Moulding

Segment



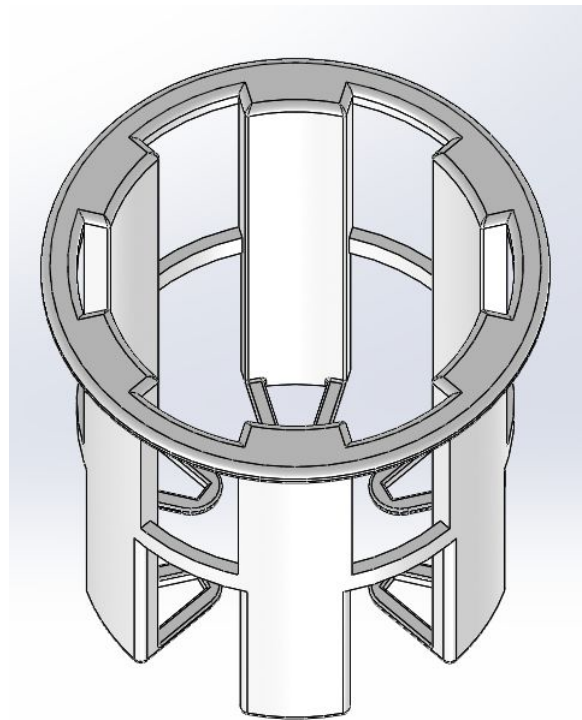
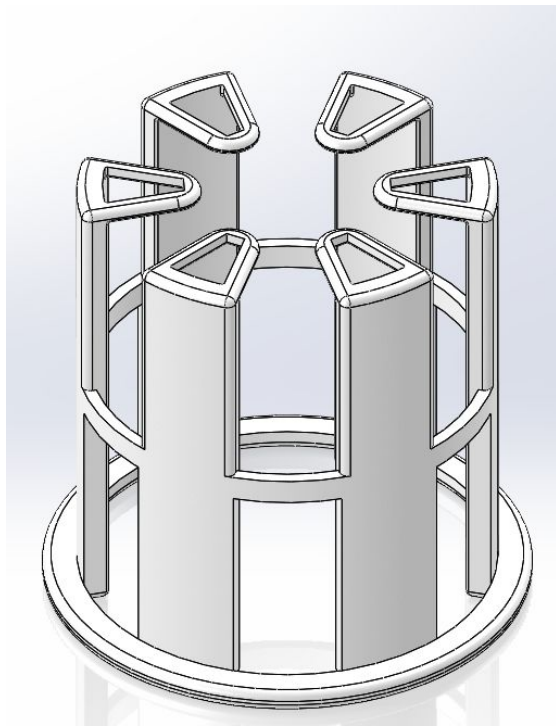
Segment

Mass = 181 grams

Material = ABS

Manufacturing method = Injection Moulding

Pod



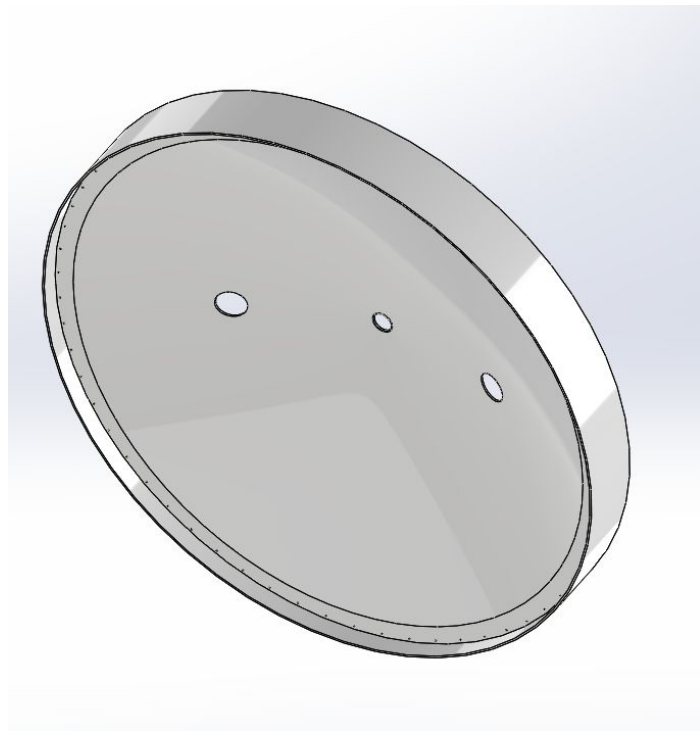
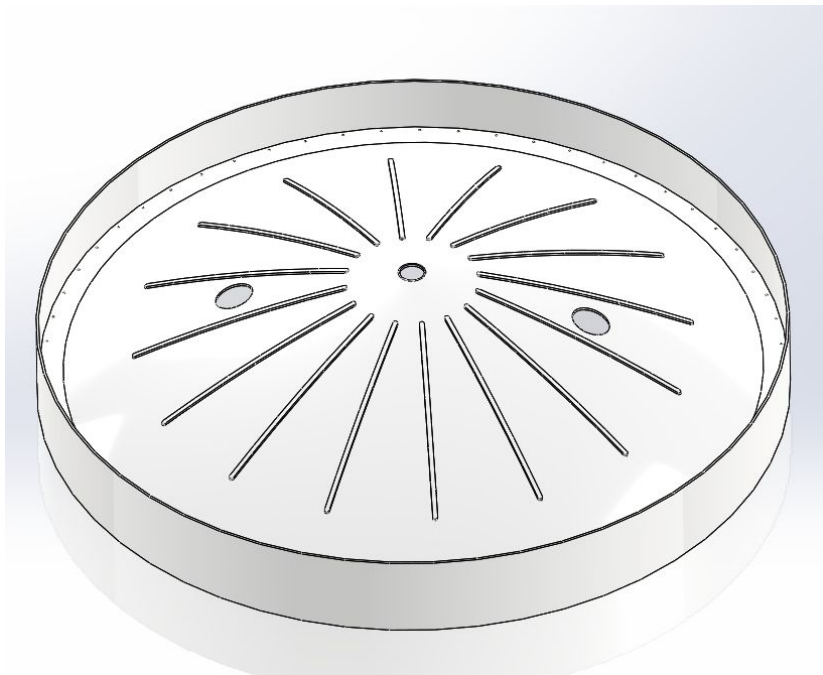
Growing pod

Mass = 15 grams

Material = ABS

Manufacturing method = Injection Moulding

Water Buffer



Water Buffer

Mass

= 1172 grams

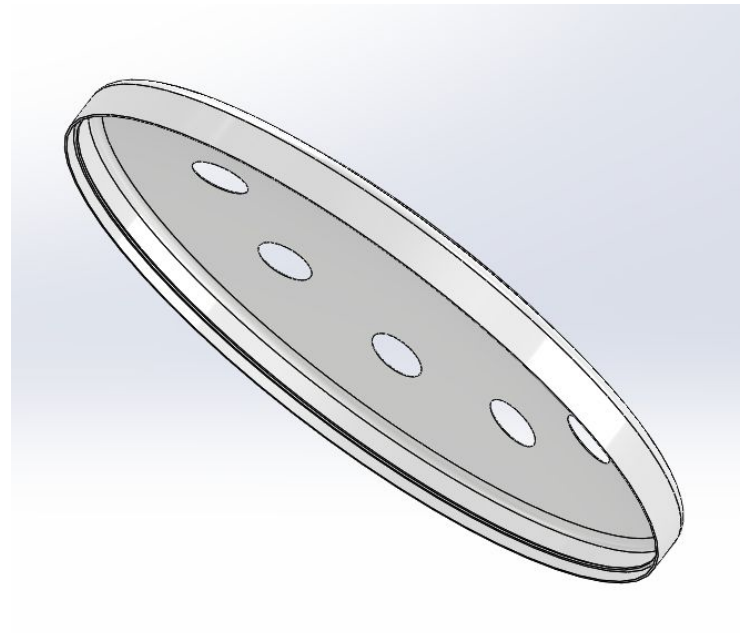
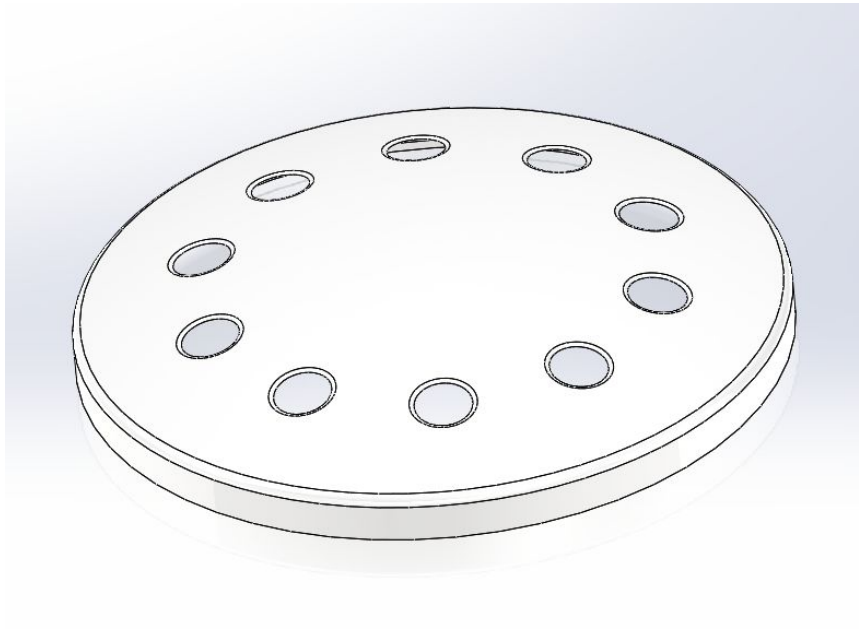
Material

= PVC Sheet

Manufacturing method

= Vacuum Moulding

Lid



Lid

Mass = 935 grams

Material = PVC Sheet

Manufacturing method = Vacuum Moulding

Pre-Fabricated parts

Tube

Aquamarine 2 Meters Aquarium Airline Tubing Soft Air Pump
Hose for Fish Tank Accessories Standard Diameter 6mm Pet
Supplies



Pump

Elove 9 Watt Water Lifting Submersible Pump for
Aquarium, Fountains - 165-250AC/50 Hz, 0.75M



Valve

Venus Aqua Outlet Tube Adjustable Connector, Air Pump Flow Control Valves Switch Tap 4x6mm Hose, Aquarium Air Tube Outlet Multicolor Connector 24 Pcs (All in One Aquarium Air Valve Combo)



Nutrient Solution

GreenLoop Hydroponic Nutrients-LEAFY-200 Makes 200 Liter hydroponic Solution. Complete with All nutrients. Suitable for All Plants & Better Leafies.



Connector

Lukzer 2 PC Garden Water Hose/One Touch Joint/Garden Water Hose Connector (Random Colour)





Farming made easier than ever before !

Thank You,

Name

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